AGILE TEAM CHARTER

WaterWatch — Smart Water Monitoring System

PRP370S – Project Presentation III | Cape Peninsula University of Technology

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Team Purpose

Objectives of the team:

The team's primary mission is to develop WaterWatch, a smart water monitoring system that solves the lack of real-time water usage visibility in Cape Town households. The project goes beyond product development – it aims to address a genuine community problem affecting homes in water-restricted areas.

The initiative is driven by research showing that most home owners have no clear insight into their daily water usage, resulting n undetected leaks, unexpectedly high bills, and unnecessary water usage.

Vision Statement

To create a system that empowers households to make informed choices about their water usage through accessible, real-time data – ultimately driving environmentally friendly consumption and positive behavioural changes at community levels.

Definition of Success

- Clear linkage between the identified problem and the stated solution.
- Delivery of a practical, technically feasible, and context appropriate prototype.
- Thorough demonstration of the problem space through research and market validation.
- Strong coherence across all deliverables (research, design, and technical solution).

Product Responsibility

- Design a non-invasive, clamp-on ultrasonic sensor capable of real-time water flow measurement.

- Develop an intuitive mobile application that converts sensor data into actionable insights such as real-time monitoring, leak alerts and bill predictions.
- Demonstrate seamless interaction between the hardware sensor and the mobile application.

2. Team Roles and Responsibilities

Name	Agile Role	Responsibilities
Matimu Mabunda	Project Manager	Facilitates team processes, coordinates the work, tracks progress using task boards, removes blockers, and maintains team communication and collaboration
Kamvelihle Mqadi	Research Analyst	Conducts research, defines and prioritises requirements, ensures the product meets the user needs, maintains the product backlog, and validates ideas with data.
Marco Julian Human	Frontend Developer and UI Designer	Designs and develops the mobile app user interface, ensures accessibility and usability, creates interactive prototypes, and collaborates the backend developer on integration.
Thapelo Masingi	Backend Developer	Handles backend architecture and logic, manages data acquisition, processing and storage, develops sensor to app integration, and ensures performance and security.

3. Team Values and Principles

3.1 Core Values:

- Accountability – Each member takes full ownership of their responsibilities and deadlines, and proactively communicates any challenges.
- Transparency – Progress, challenges and updates are openly shared through team tools and platforms.
- Respect – All opinions, skills and contributions are valued through active listening and constructive feedback.
- Reliability – Team members consistently meet deadlines and deliver quality work to maintain a smooth workflow.
- Continuous Improvement – Regular reflection and feedback are used to enhance processes and outcomes.
- Commitment to Quality – All deliverables meet high academic and professional standards.

3.2 Application of Agile Principles

- Incremental Delivery – Work is broken into small, manageable tasks that are regularly completed and reviewed.
- Continuous Feedback Integration – Feedback is actively sought and applied to improve designs and resolve issues early.
- Adaptability and Responsiveness – The team remains flexible and adjusts plans basked on new insights or challenges.
- Collaborative Working Approach – Regular communication and knowledge sharing support collective decision making.
- User-Centred Focus – All decisions are guided by user needs identified during research.

4. Communication Plan

Agile Framework

The team follows an Agile approach using a digital task board with the following columns:

Product Backlog → In progress → Review →Completed

Communication and Task Coordination Platform

- Primary: WhatsApp for daily updates and quick problem-solving
- Secondary: In person meetings on campus for deeper collaboration and reviews.'

Frequency of Meetings

Ceremony	Frequency	Purpose
Stand-up	15 Minutes, Weekdays	Track daily progress and resolve blockers
Spring Planning	Weekly	Define tasks and priorities for the sprint
Sprint Review	Weekends	Evaluate completed work
Retrospective	Monday and Friday	Reflect on teamwork and processes.

Working Agreements

- Attend meetings prepared and on time
- Proactively inform the team about potential delays
- Participate actively in discussions and decisions
- Respect different working styles and viewpoints
- Focus on Solutions rather than problems.

Goals and Milestones

SMART goals

Criteria	Goal
Specific	Design and Develop a smart water monitoring system that provides real-time visibility and leak detection for Cape Town.
Measurable	Successfully deliver all project milestones including research, design, and functional prototype.
Achievable	Leverage the team's combines skills in research, design, frontend and backend development

Relevant	Directly address the validated problem of water usage invincibility
Time bound	Complete the full project within the 9-week timeframe

Project Milestones:

Milestone	Timeline
Research and Validation of Problem	Week 2
Product Concept and Requirements Defined	Week 4
Design and Prototype	Week 6
Review and Refinement of Deliverables	Week 8
Final Submission	Week 9

Risks, Constraints, and Mitigation Strategies

Category	Item	Mitigation Strategy
Risk	Unequal contribution	Regular stand-ups and visible Kanban board
Risk	Miscommunication or delays	Clear agreements and early flagging of issues
Risk	Lack of technical experience	Peer learning, strength-based task allocation, and lecturer support
Constraint	Completing module workloads	Careful sprint planning and workload balancing
Constraint	Limited in-person meeting time	Use of asynchronous updates via WhatsApp
Constraint	No Physical hardware prototype	Focus on high-fidelity digital prototypes and detailed documentation.